Andy(Jixin) Li

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EDUCATION

Carnegie Mellon University *Expected Graduation: May 2028*
Master of Computational Data Science (MCDS)
University of Illinois Urbana-Champaign (UIUC) *Sep 2022 - May 2026*
Bachelor of Science in Computer Science & Economics Cumulative GPA: 3.96/4.00

SKILLS

Programming & Frameworks: Python, C++, Java, C#, JavaScript, TypeScript, Go, SQL, Kotlin
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, React, Node.js, Spring Boot, Flask
Systems & Developer Tools: Linux/Unix, Git, Docker, Bash, CI/CD, REST APIs, Web Services, Microservices, Multithreading, Concurrent Programming, Distributed Systems, Distributed Data Processing
AI & Data: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Retrieval-Augmented Generation, Information Retrieval, Active Learning, Model Evaluation, Database Systems
Software Engineering: Backend Development, Full-Stack Development, Data Structures and Algorithms, Software Testing, System Design, Design Patterns, Agile Development, Software Development Life Cycle, Troubleshooting

WORK EXPERIENCE

Intel Corporation | ***AI Software Engineer Graduate Intern*** | Hillsboro, OR *June 2026 – August 2026*
Developed an AI-assisted industrial defect annotation and evaluation framework using Intel Geti Active Learning

- Evaluated Geti's AI-assisted annotation workflow against manual labeling and traditional **YOLO** workflow, **reducing annotation time by over 50%** while maintaining **90% annotation accuracy** across multiple industrial defect categories
- Designed prompt engineering pipelines to benchmark AI-assisted annotation strategies under different confidence thresholds, reducing the **precision–recall gap** by approximately **30%** and mitigating model prediction bias
- Authored Intel's internal benchmarking documentation, establishing standardized evaluation guidelines adopted by 50+ engineers for AI-assisted annotation workflows

National Center for Supercomputing Applications | ***AI Software Engineer Intern*** | Champaign, IL *Jan 2025 – March 2026*
Project (i): AI-Agribench (AI-FARM): *Developed a benchmark for agricultural LLM evaluations for NCSA*

- Developed a **multi-stage filtering pipeline** to extract around 5,000 high-quality question–answer pairs from 399,199 Extension publications, using rule-based filtering and semantic filtering with all-mpnet-base-v2 embeddings across 8 agronomic categories
- Utilized a GPT-4o-mini-driven **QA generation workflow** "yourbench" to produce long-form answers with inline citations and uniformly sample instances for evaluation to domain experts, ensuring practical decision-support in commercial farming contexts
- Established an "LLM-as-a-Judge" **evaluation framework** using GPT-OSS-120b to assess models on Accuracy, Relevance, Completeness, and Parsimony, identifying performance gaps and informing future benchmark iterations
- Set up leaderboard website for daily usage purpose and receive **Gates Foundation** for future development

Project (ii): Feat2Spec (SPIN Program): *Built a LLM–driven framework for spectrum generation and soil prediction*

- Utilized **SOGM**, a text-conditioned **denoising diffusion model**, to simulate soil reflectance (400–2499 nm) from 13-dimensional physical and chemical property inputs, achieving 5.5% RMSE ($R^2 = 0.86$) on unseen spectra
- Designed a multithreaded **CNN** to inversely predict soil from 2151-band spectral vectors for high-precision property inference
- Integrated the **CNN–SOGM dual models** into **LLaMA 3** within a "**Proposer → Verifier → Executor**" reasoning architecture for interpretable and physically consistent predictions
- Built a **text-based interface** allowing users to describe soil conditions in natural language and generate corresponding reflectance–property pairs, solving previous performance bottlenecks in SOGM

Agricultural Bank of China | ***AI Software Engineer Intern*** | Remote *Feb 2025 – May 2025*
*Developed a document understanding system for financial PDFs using **multi-modal LLMs***

- Fine-tuned multi-modal models (**LayoutLMv3, Donut**) on internal financial reports to improve structured data extraction
- Integrated **PaddleOCR** with **CRF-based** token alignment to enhance entity segmentation and alignment across multi-page reports
- Built an annotation loop with **Doccano**, integrating weak labels and manual corrections, and adapting to bank-specific formats
- Designed a **RAG-inspired pipeline** for downstream use cases, including semantic retrieval, report summarization, and cross-document linkage, contributing to internal knowledge mining tools

SeeU International Inc. | ***AI Software Engineer Intern*** | *Remote* *Dec 2024 – Feb 2025*
*Engineered a job recommendation system with **LLMs (GPT-3.5/4)** and **cosine similarity** to automate resume-job matching, reducing manual review time by 40+% for 500+ student users*

- Built a **Python** pipeline to process unstructured resume/JD data into structured **JSON** embeddings with DeepSeek and OpenAI APIs, handling 1,000+ documents weekly and used **Agile/Scrum**, deploying updates via **Git** and **Docker** for CI/CD